Sushank Singh

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Profile

Computer Science student at VIT-AP University with a strong academic record (CGPA: 9.16) and a passion for backend and full-stack software engineering.
Experienced in designing scalable REST APIs, authentication systems, analytics pipelines, caching, background task processing, and modern deployment workflows through production-style projects.
Strong foundation in Data Structures and Algorithms with hands-on experience building real-world applications using modern web technologies.

Education

VIT-AP University — Amaravati, Andhra Pradesh 2024 – 2028
B.Tech. in Computer Science and Engineering CGPA: 9.16

Kendriya Vidyalaya — Ayodhya, Uttar Pradesh

- Class XII: 86%
- Class X: 91%

Technologies

- **Languages:** Python, C++, Java, SQL, HTML5, CSS3, JavaScript
- **Backend:** FastAPI, Flask, SQLAlchemy, Alembic, REST APIs, JWT Authentication
- **Frontend:** React, Vite, Tailwind CSS, Axios, React Router
- **Databases:** PostgreSQL, MySQL, SQLite
- **DevOps & Tools:** Docker, Docker Compose, Redis, Celery, Git, GitHub, GitHub Actions, Render
- **Testing:** Pytest, FastAPI Dependency Overrides, Unit Testing, Integration Testing
- **Machine Learning:** Scikit-learn, Pandas, NumPy, Feature Engineering, Model Evaluation
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

Projects

CP Analytics – AI-Powered Competitive Programming Analytics Platform GitHub — Live — Demo
Full-Stack Developer

- Built a production-style competitive programming analytics platform that synchronizes Codeforces profiles and generates actionable insights.
- Implemented JWT authentication, REST APIs, SQLAlchemy ORM, Alembic migrations, and a modular FastAPI backend with PostgreSQL.
- Designed analytics for contest performance, daily activity, tag statistics, recommendations, and upsolve tracking.
- Integrated Redis caching, Celery, Docker, GitHub Actions, Pytest, and deployed the application on Render.

SafeMap – Safety-Aware Route Recommendation Platform GitHub
AI/ML & Backend Contributor — Hackathon Winner

- Developed an AI-powered safety-aware navigation platform using real-time hazard detection.
- Built an emergency siren detection model and backend APIs for intelligent route optimization.
- Integrated pothole detection with backend services to improve routing accuracy and user safety.

SafeShip – AI-Driven Package Damage Detection GitHub
Backend Developer — Hackathon Winner

- Developed FFT-based REST APIs for package damage detection using audio signal processing.
- Designed backend pipelines for anomaly detection and inspection result generation.
- Integrated AI models, frontend, and backend services for real-time package monitoring.

Achievements

- National Level Hackathon AI/ML 1st Rank - SRM AP
- Hackathon Winner – D3 Fest, IIIT Bhubaneswar
- Hackathon Winner – THE MATRIX, VIT-AP
- CS50x (edX) – Completed: June 2025
- Solved 350+ DSA problems across platforms like LeetCode, CodeChef, and GeeksforGeeks
- Participated in 15+ competitive programming contests